

1 Oxygen is needed for aerobic respiration in cells.

In eukaryotic cells, the mitochondrion is the organelle used for aerobic respiration.

(a) Fig. 1.1 shows a transmission electron micrograph of a mitochondrion.

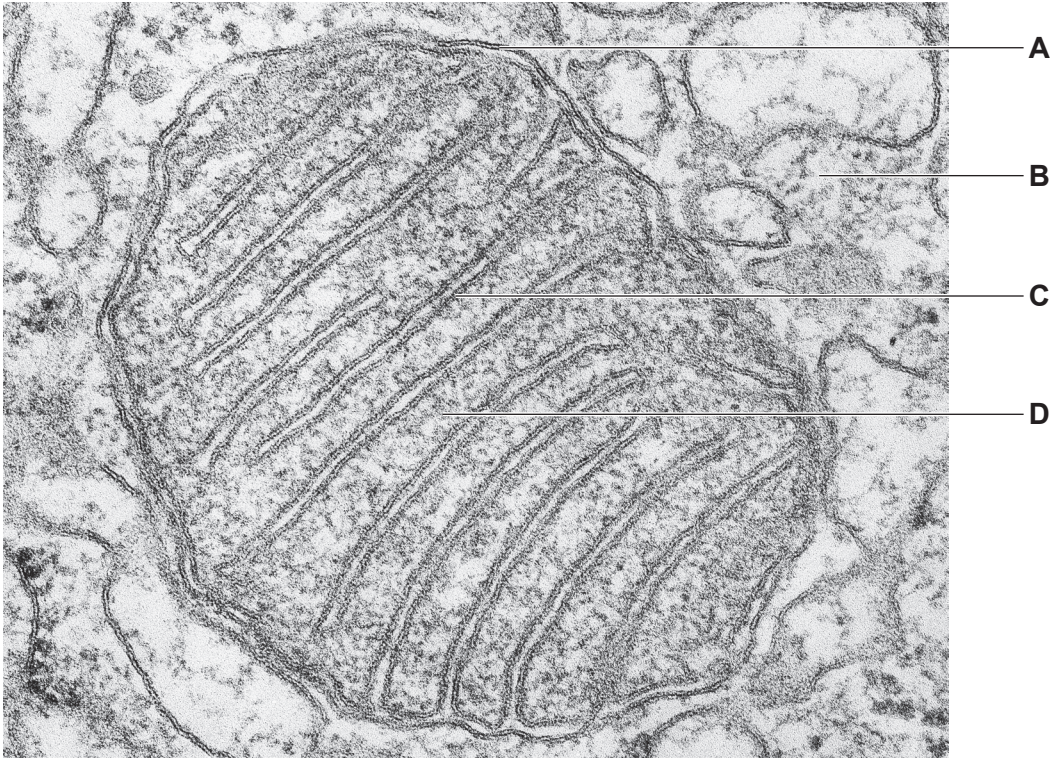


Fig. 1.1

Table 1.1 lists the four stages of aerobic respiration.

With reference to Fig. 1.1, complete Table 1.1 using the letters **A–D** to show where each stage occurs.

Each letter may be used once, more than once or not at all.

Table 1.1

stage of respiration	letter
glycolysis	
link reaction	
Krebs cycle	
oxidative phosphorylation	

- (b) In anaerobic conditions, no ATP can be synthesised by oxidative phosphorylation because the process stops.

Explain why ATP synthesis by oxidative phosphorylation stops in anaerobic conditions.

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- (c) Up to 60% of the ATP that is produced in cancer cells comes from lactate fermentation of glucose, even though oxygen is present.

Scientists are developing cancer treatments to inhibit the enzyme that catalyses the last step of lactate fermentation.

Explain how the inhibition of this enzyme reduces the production of ATP in cancer cells.

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